NEXUS CONTROL

Master Product, UX and Implementation Specification

Klipper + Moonraker Cybernetic Touch Console



High-fidelity concept direction. Production UI uses original assets and the capability-aware controls defined in this document.

PACKAGE PURPOSE	Implementation handoff for a coding AI or software team. Includes visual references, page contracts, design tokens, API map, safety model and acceptance criteria.

Working concept: an original cybernetic, Borg-inspired Klipper/Moonraker touchscreen interface that looks sophisticated and intimidatingly capable while remaining readable, predictable and safe for daily 3D-printing work. The package intentionally avoids Star Trek logos, Borg insignia or copied screen graphics; it uses an original visual system built from dark graphite surfaces, emerald circuitry, cut-corner panel geometry and precise technical typography.

This document is written as an implementation contract for a coding AI or software team. When a visual choice and a safety/data-integrity rule conflict, the safety/data-integrity rule wins.

1. Product goals

- Feel like a purpose-built machine console, not a generic website reskinned green.
- Allow a specialist to understand the machine in seconds, yet keep advanced tuning and diagnostics one tap away.
- Use text and icons together. No important action is icon-only.
- Use live data and WebSocket subscriptions so the display feels alive without excessive polling.
- Expose only capabilities that actually exist on the connected Moonraker/Klipper system.
- Never silently alter printer.cfg, Moonraker configuration, heater safety, stepper current or persistent calibration values.
- Support touch-first operation on an x86-connected display as well as desktop/mouse use and normal browser access.
- Keep the whole app inside the viewport. The shell itself should not require browser-page scrolling; individual panels and lists may scroll internally.
- Scale from compact 800x480/1024x600 class panels to 1280x800, 1920x1080 and ultrawide operator displays.
- Support multiple printers without turning the interface into a fleet-management product. The currently selected printer is always obvious.

2. Non-negotiable UX principles

- Normal/active = emerald green, information = cyan, attention = amber, destructive/fault = red, unavailable = gray. Always pair color with text/icon state.
- Primary operational numbers use tabular numerals and are visually larger than labels.
- Dangerous operations are visually isolated. Cancel, delete, power-off, reboot and shutdown never sit next to ordinary navigation without separation.
- Emergency Stop remains globally reachable and is immediate. It does not use a delayed G-code queue path.
- A disconnected UI becomes read-only until live state is trustworthy again.
- Optional features disappear or explain their missing capability; the UI must never present a fake control that does nothing.
- Advanced pages can be dense, but controls still meet the minimum touch target.
- Every wizard has Cancel/Abort and a visible current step. Persistent writes show exactly what will change.

3. Global shell and navigation

The application shell is persistent. Only the control plane changes. This gives the interface the feeling of a physical console with selectable planes rather than a conventional website.

- Top status rail: product name, current printer selector, connection state, Klippy state, active print mini-status, warning count and clock.
- Left control-plane rail grouped as STATUS, CONTROL, CALIBRATE, AUTOMATE, OBSERVE and SYSTEM.
- Expanded sidebar shows icon + text. Compact mode shows a large glyph and short tooltip/long-press label.
- Selected plane has an illuminated emerald edge, slightly brighter panel background and an active glyph. Do not use a huge glow that reduces text contrast.
- Page header contains title, one-sentence purpose and local sub-tabs.
- Global quick drawer can be opened from top bar for Emergency Stop, Pause/Resume, Park, Home and Cooldown. It must not obscure an active critical alert.
- Persistent fault rail appears above page content when Klipper is shutdown/error, Moonraker reports failed components, or a high-priority announcement requires attention.

4. Visual language

The intended look is dark, cybernetic and machine-like. It should remind a viewer of a sophisticated collective/biomechanical computer without using protected franchise marks. The visual treatment must remain subordinate to operational clarity.

- Background: near-black green/graphite with extremely subtle grid/circuit texture.

- Panels: dark green-black with 1 px green structural border, cut/segmented corners where practical, and restrained inner glow.
- Accent: #00F28A emerald. Bright accent: #68FFC0. Cyan reserved for secondary sensor/information channels.
- Amber and red are reserved for state semantics, not decoration.
- Charts use thin luminous lines on quiet grids. Avoid filled neon areas behind dense text.
- Use ambient trace animations at low opacity. Reduced Motion turns them off.
- No blinking normal-state LEDs. Flashing is reserved for a user-configurable critical alarm and must stop when acknowledged.
- Icons are simple original line glyphs. Pair every important icon with text.
- Minimum normal touch target 48x48 px; preferred kiosk target 56x56 px.
- All numerical control widgets support direct keypad entry as well as sliders/chips where appropriate.

5. Responsive behavior

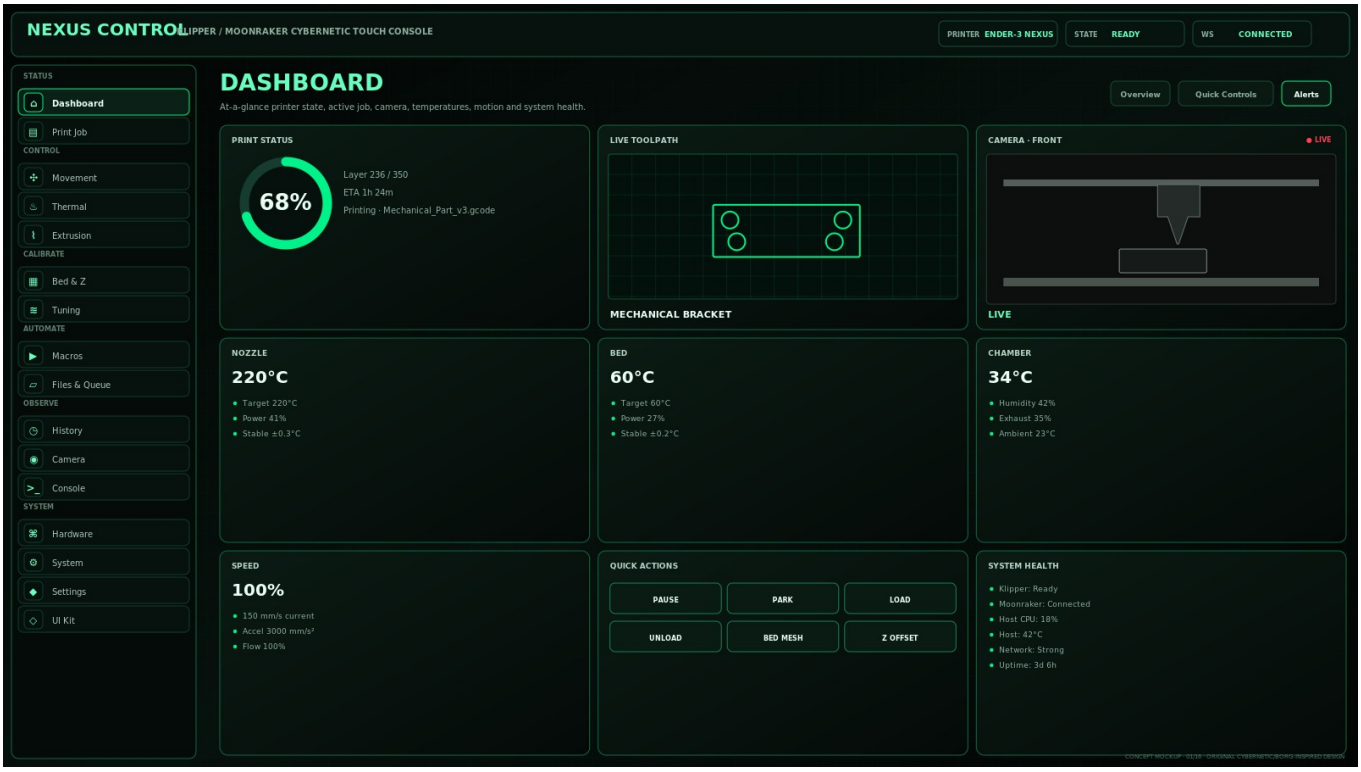
- 800-1023 px width: compact sidebar, one or two content columns, larger modal sheets, fewer simultaneous chart labels.
- 1024-1279 px: compact or expanded sidebar user choice, two content columns.
- 1280-1599 px: standard three-column dashboard grid.
- 1600+ px: three/four-column grid with camera/toolpath hero panels allowed to span two columns.
- Ultrawide: never stretch text lines excessively; increase columns/panel grouping instead.
- Viewport height is authoritative. The top rail and sidebar stay fixed. Data-heavy lists, terminal output and file lists scroll inside their own panel.
- Landscape is the primary control-panel orientation. Portrait remains functional but is not the primary visual target.

6. Live-data architecture

- Use Moonraker HTTP for discovery, one-off commands and transfers; use WebSocket JSON-RPC subscriptions for ongoing printer state.
- On connect: read server info, printer info, loaded printer objects, then build the subscription dynamically.
- Normalize partial WebSocket updates into one typed state store. Do not replace missing fields with zero; absence and zero are different states.
- Every state field tracks received timestamp and stale status.
- After disconnect: mark live values stale, stop controls that can change the printer, show RECONNECTING, then exponential backoff 1/2/4/8/16/30 seconds.
- If a limited polling fallback is needed, use a 1-2 second cadence for core print state rather than aggressive sub-second polling.
- Charts maintain a bounded client-side ring buffer. Long-term analytics use Moonraker history or an optional external data source rather than unbounded browser memory.
- Capability discovery is repeated after Klipper restart, firmware restart, config reload or version/config fingerprint change.

7. Control planes - complete specification

7.1 Dashboard



Control-plane mockup: dashboard. Editable SVG is included in the package.

The landing plane. A user should understand printer state, active job, temperatures, camera, urgent warnings and the next useful action within two seconds.

Primary sub-tabs: Overview, Quick Controls, Alerts.

Layout

- Top-left: printer state and circular print progress.
- Top-center: current G-code model/toolpath preview and file identity.
- Top-right: selected live camera.
- Middle row: nozzle, bed, chamber, fan, speed and flow cards.
- Bottom-left: condensed movement widget.
- Bottom-center: favorite macros / quick actions.
- Bottom-right: Klipper, Moonraker, host, network and uptime health.

Options and controls

- Printer selector for multi-printer installations.
- Connection indicator with Connected / Reconnecting / Offline / Stale states.
- Printer state badge: ready, printing, paused, complete, error, shutdown, startup.
- Print progress with percentage, current layer, total layer when metadata exists, elapsed time and estimated remaining time.
- Current file name, folder, thumbnail and slicer metadata.
- Optional 2D/3D toolpath preview.
- Camera card with user-selected default camera, full-screen button and snapshot button.
- Nozzle cards for every detected extruder, not only extruder0.
- Bed heater card when heater_bed exists.
- Chamber / ambient / MCU / host temperature cards when those sensors exist.
- Part fan and optional auxiliary fan quick controls.
- Speed factor and flow factor readouts with one-tap reset.
- Quick pause/resume/cancel controls while a job is active.
- Quick Home All, Park, Load, Unload, Bed Mesh and Z Offset actions when valid.
- Favorite macros chosen by the user.

- Warning rail for Klipper errors, Moonraker warnings, sensor faults and high-priority announcements.
- System health card: Klipper state, Moonraker state, host CPU, host memory, host temperature, network activity, storage warning, uptime.
- Optional spool card showing active Spoolman spool and remaining material.
- Widget edit mode: reorder, hide, resize and restore defaults.
- Kiosk mode can hide editing affordances and expand live controls.

Data / capability sources

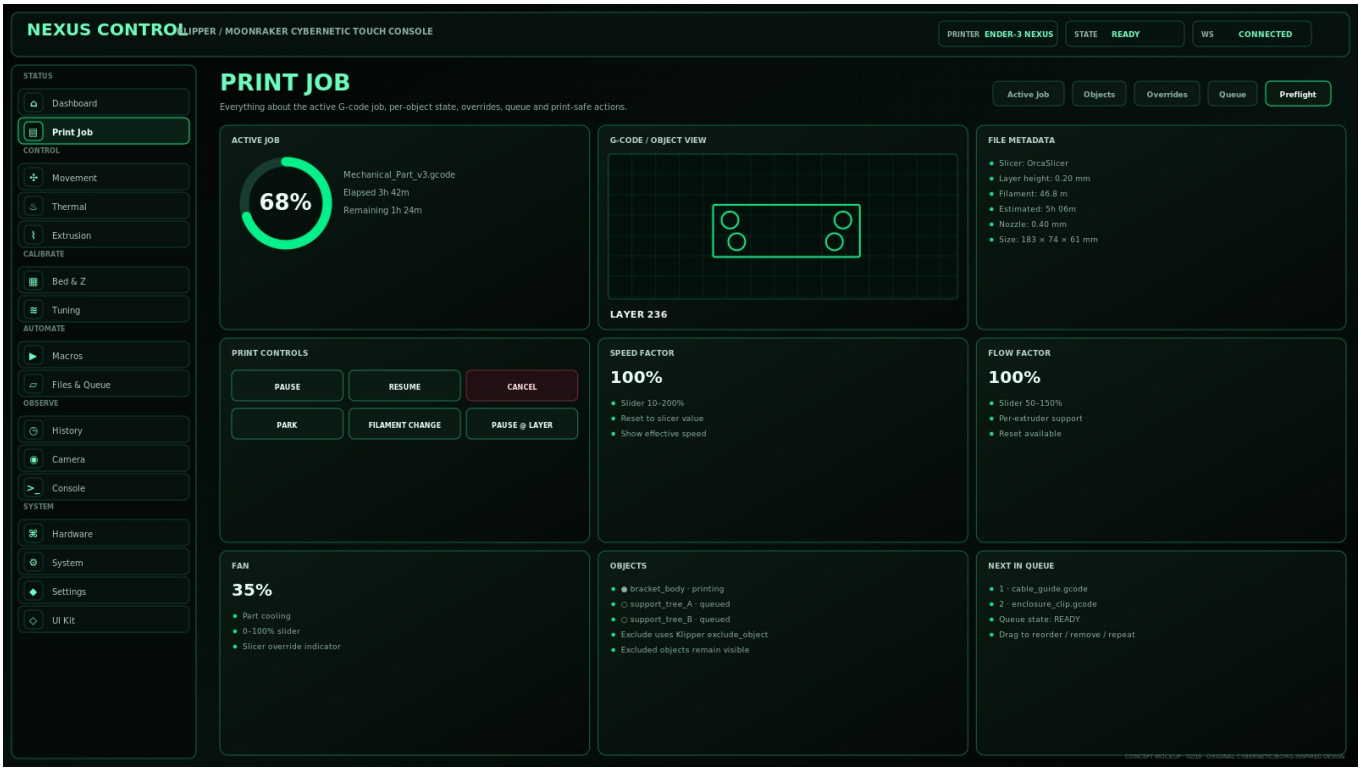
- printer.info / server.info
- printer objects: webhooks, print_stats, virtual_sdcard, display_status, toolhead, gcode_move, extruder*, heater_bed, fan*, temperature_sensor*, system_stats where available
- Moonraker process-stat notifications
- webcam list
- announcements
- Spoolman when configured

Safety and behavior

- Cancel requires confirmation.
- Emergency stop is globally accessible but visually separated from normal print controls.
- All action buttons disable when connection is stale or required Klipper state is not ready.

Visual reference: images/01_dashboard.png and editable vector images/01_dashboard.svg.

7.2 Print Job



Control-plane mockup: print. Editable SVG is included in the package.

The detailed active-job plane. It provides print progression, per-object control, live overrides, queue context and safe intervention.

Primary sub-tabs: Active Job, Objects, Overrides, Queue, Preflight.

Layout

- Hero progress block on the left.
- Large object/toolpath visualization in the center.
- Metadata and object list on the right.
- Lower panels for speed/flow/fan overrides, print actions, queue and preflight.

Options and controls

- Start selected file.
- Pause active job.
- Resume paused job.
- Cancel active job with confirmation.
- Reprint last completed file if it still exists.
- Show progress, elapsed, print duration, total duration, estimated remaining, layer, Z height and active object.
- Show G-code metadata: slicer, layer height, first layer, nozzle diameter, filament estimate, object bounds, estimated duration, thumbnails and modified date when available.
- Object list from Klipper exclude_object definitions.
- Highlight current object.
- Exclude a single object with confirmation and clear explanation that skipped geometry cannot be recovered later.
- Show already excluded objects in red/gray rather than removing them.
- Speed factor slider plus numeric keypad and reset.
- Flow factor slider plus numeric keypad and reset.
- Part cooling fan override and reset to slicer control.
- Pause-at-layer helper implemented through a safe macro / command strategy, not by pretending Klipper has a generic native scheduling UI.
- Filament-change action mapped to configured macro when available.
- Park action mapped to configured macro or explicit configured park coordinates.

- Live temperature mini-cards while printing.
- Queue preview and “add current file again” repeat action.
- Preflight summary before start: Klippy ready, homing policy, configured tool/nozzle, known filament, file fit metadata, file existence, relevant warnings.
- Optional one-time “Start anyway” only for non-critical preflight warnings.
- Current command / macro banner during a user-initiated operation.
- Job user identity when Moonraker history provides it.
- Object thumbnail click opens object details.
- Print notes / tags stored in UI database namespace, never injected into G-code.

Data / capability sources

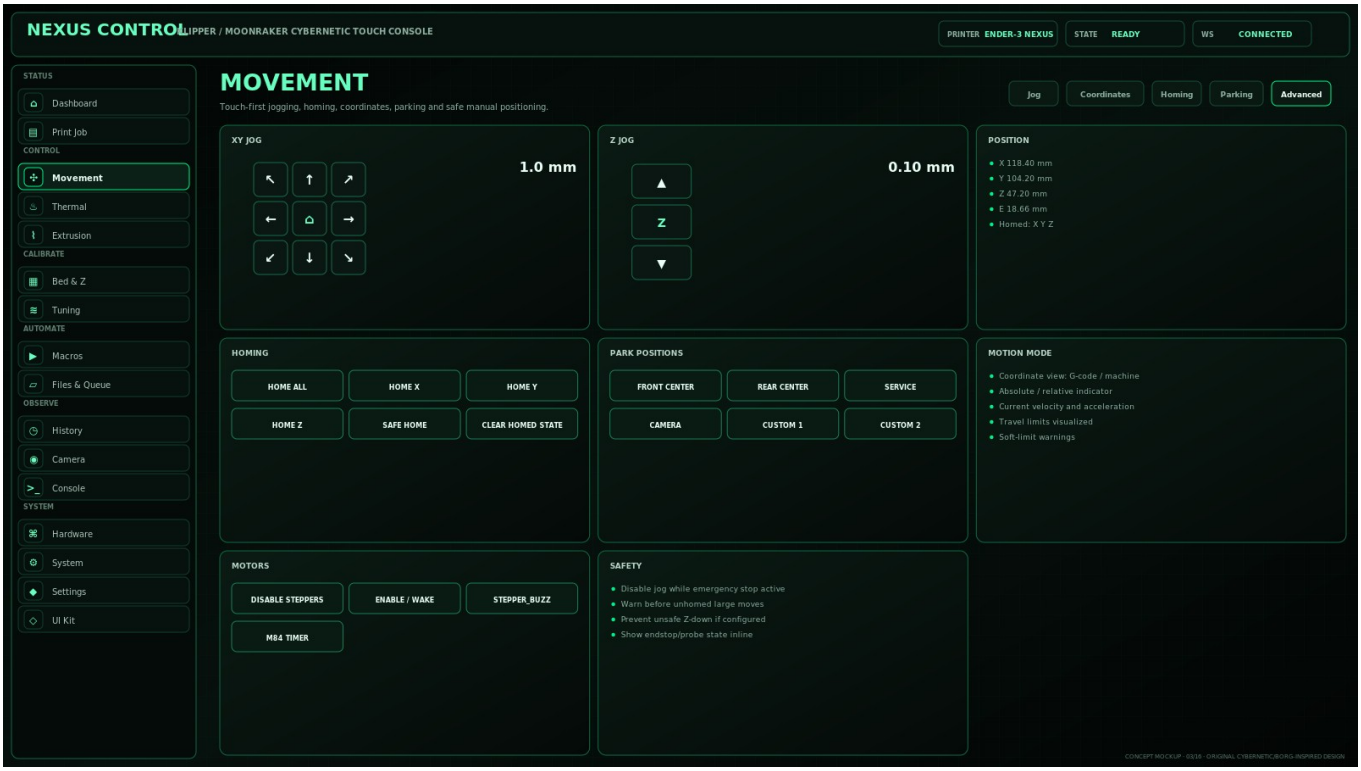
- Moonraker print start/pause/resume/cancel APIs
- print_stats, virtual_sdcard, display_status, toolhead, gcode_move
- exclude_object object
- file metadata / thumbnails
- job queue and history

Safety and behavior

- M112 entered in console must be intercepted and sent through /printer/emergency_stop for immediate halt.
- Print controls are state-aware; Start only when the printer/file is valid, Resume only when paused, Pause/Cancel only when printing or paused as appropriate.

Visual reference: images/02_print.png and editable vector images/02_print.svg.

7.3 Movement



Control-plane mockup: movement. Editable SVG is included in the package.

Manual positioning that feels like a physical machine console while preventing accidental long or unsafe moves.

Primary sub-tabs: Jog, Coordinates, Homing, Parking, Advanced.

Layout

- Large XY directional pad.
- Independent Z up/down column.
- Position and homing status.
- Step-size and jog-speed controls.
- Home, park and motor actions.

Options and controls

- XY jog in eight directions or cardinal-only preference.
- Z up and Z down.
- Jog increments: 0.01, 0.05, 0.1, 1, 5, 10, 25, 50, 100 mm; user can choose which chips appear.
- Separate Z increment set with smaller defaults.
- Jog speed selector, with separate XY and Z defaults.
- Hold-to-repeat as an opt-in touch feature with acceleration disabled by default.
- Home All, Home X, Home Y and Home Z.
- Safe-home macro button when configured.
- Display X/Y/Z/E positions and homed_axes.
- Toggle G-code coordinates vs machine coordinates when data supports it.
- Absolute/relative mode indicator; do not silently switch modes for console state.
- Configurable park positions with names, icons and safe-Z requirements.
- Disable steppers with confirmation when Z could fall under gravity; warning text can be printer-profile specific.
- Stepper buzz diagnostic action only under Advanced.
- Endstop status drawer / QUERY_ENDSTOPS.
- Probe state indicator when a probe exists.
- Soft-limit visualization based on config limits.
- Optional bed/nozzle miniature map showing current XY location.

- User-defined “service position” and “camera position.”

Data / capability sources

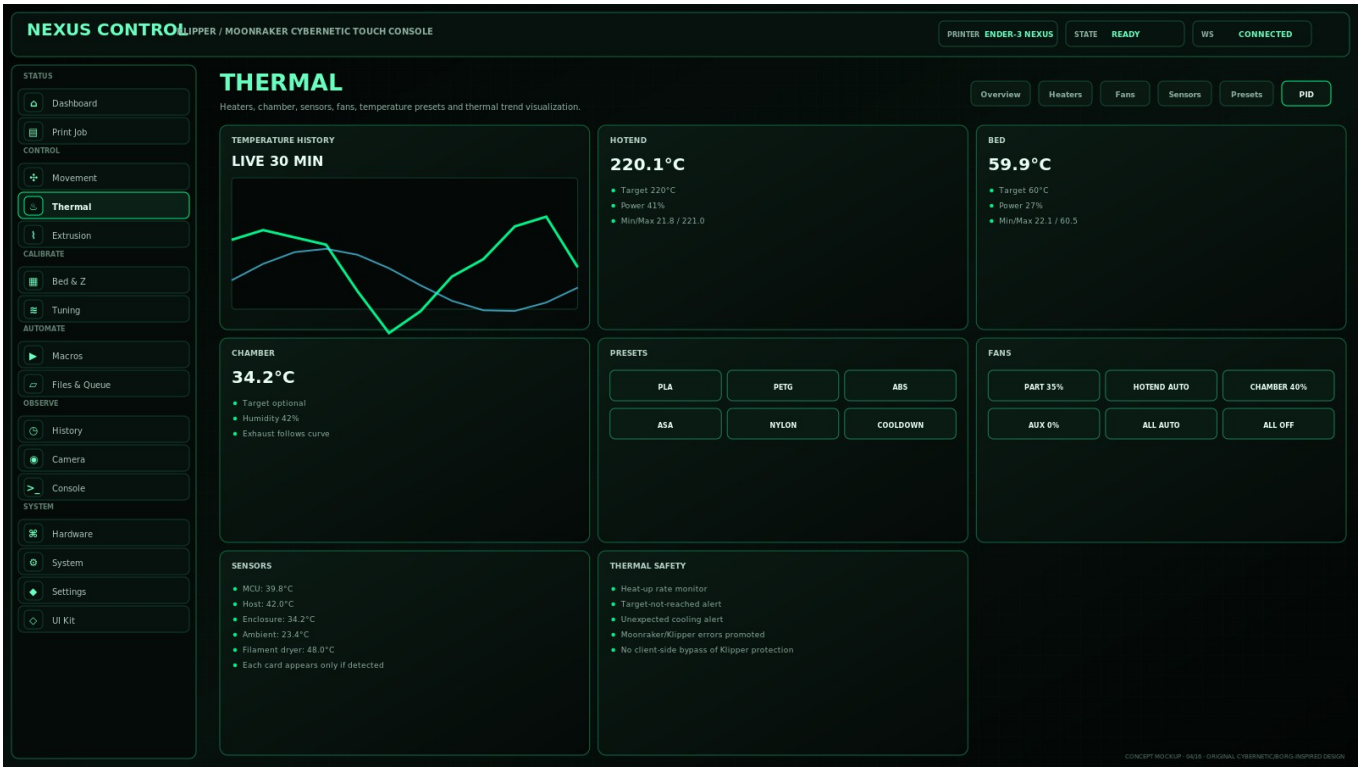
- toolhead, gcode_move, stepper_enable where available
- printer.endstops.status or QUERY_ENDSTOPS
- G-code scripts for G28 and controlled G0/G1 jogging

Safety and behavior

- Large moves require homed axes unless user explicitly uses a configured recovery profile.
- Z-down controls may show a near-bed caution zone.
- Disable motors never appears as a bright primary action.

Visual reference: images/03_movement.png and editable vector images/03_movement.svg.

7.4 Thermal



Control-plane mockup: thermal. Editable SVG is included in the package.

A unified heater, sensor and fan plane with excellent trend visibility and no attempt to replace Klipper thermal protection.

Primary sub-tabs: Overview, Heaters, Fans, Sensors, Presets, PID.

Layout

- Large multi-series temperature chart.
- Heater cards.
- Fan cards.
- Sensor list.
- Preset and PID tabs.

Options and controls

- Live chart for every detected heater and selected temperature sensors.
- Chart windows: 5 min, 30 min, 2 h, current job and custom.
- Target lines and optional heater-power overlay.
- Nozzle target entry for each extruder.
- Bed target entry.
- Chamber heater target if configured.
- Cooldown all heaters.
- Temperature presets: PLA, PETG, ABS, ASA, Nylon, TPU plus user-defined profiles.
- Preset profile can contain nozzle, bed, chamber and fan values.
- Part fan slider, controller fan state, heater-fan state, temperature-fan target/state and generic-fan control only when those are controllable.
- Tachometer RPM shown only when provided.
- Sensor min/max since connection, current reading and stale timestamp.
- Host and MCU temperature cards if available.
- Heater power 0–100% visualization.
- Heat-up timer and rate-of-change indicator.
- PID tuning entry point, with target temperature, cycles / command details and a result review workflow.
- Pinned sensors for dashboard.

- Chart export to CSV for diagnostic sessions.
- Optional °F display conversion; all commands and persisted Klipper values remain in °C.

Data / capability sources

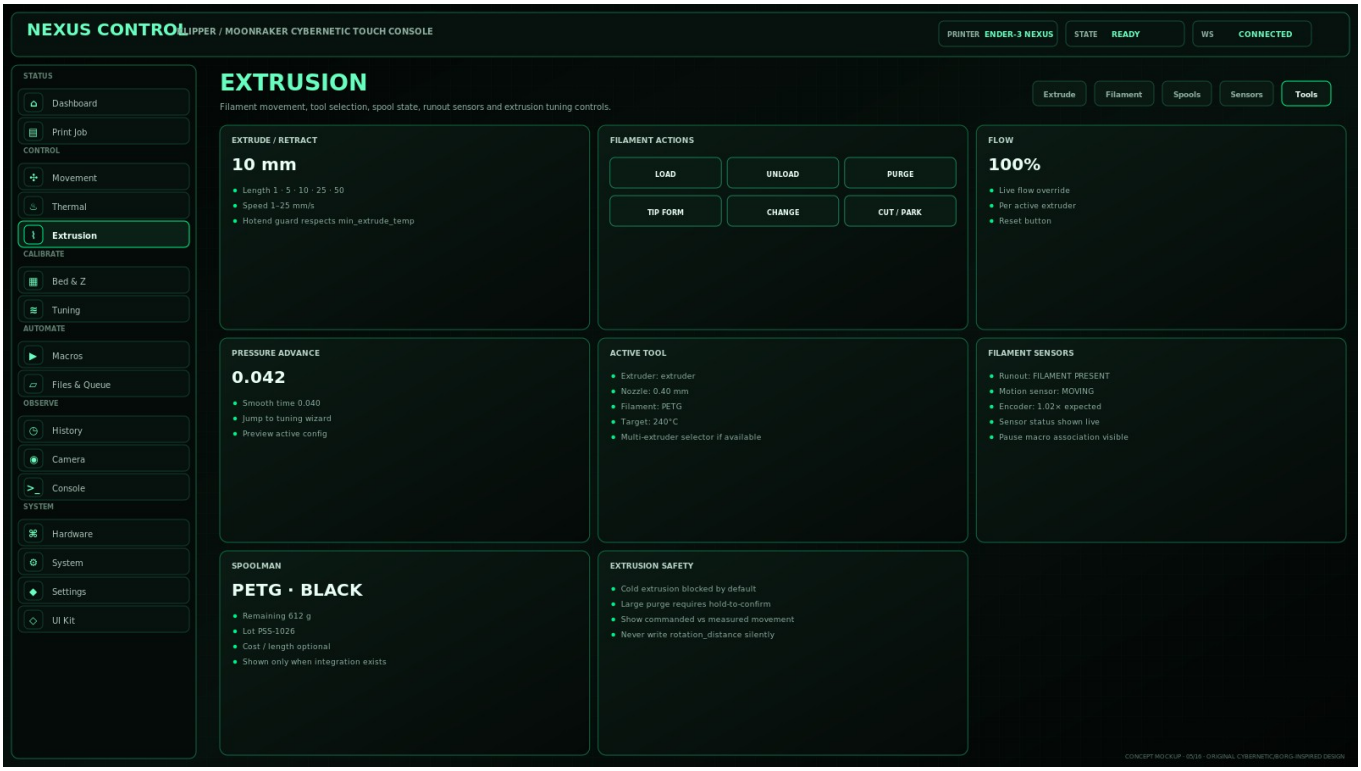
- extruder*, heater_bed, heaters, temperature_sensor*, temperature_fan*, fan/heater_fan/controller_fan/generic_fan
- Moonraker sensor component when configured
- proc-stat CPU temperature

Safety and behavior

- Never expose a client-side switch that disables verify_heater or other Klipper protections.
- Numeric targets enforce configured min/max bounds where available and require confirmation for unusually high user-defined thresholds.

Visual reference: images/04_thermal.png and editable vector images/04_thermal.svg.

7.5 Extrusion



Control-plane mockup: extrusion. Editable SVG is included in the package.

Filament handling, tool selection, runout/motion sensor visibility and normal extrusion overrides.

Primary sub-tabs: Extrude, Filament, Spools, Sensors, Tools.

Layout

- Large extrude/retract control.
- Load/unload/purge actions.
- Active tool and spool panels.
- Filament sensor cards.
- Flow and pressure-advance summaries.

Options and controls

- Extrude and retract buttons.
- Selectable extrusion length chips.
- Selectable extrusion speed.
- Custom numeric length/speed keypad.
- Load Filament and Unload Filament macro discovery / mapping.
- Purge and tip-form actions through configured macros.
- Filament Change action through configured M600-equivalent macro if present.
- Cold-extrusion guard: UI checks hotend state and relies on Klipper min_extrude_temp.
- Flow factor control.
- Active extruder/tool selector for multi-extruder systems.
- Per-extruder target temperature and current temperature.
- Nozzle diameter informational field from printer profile / file metadata.
- Pressure Advance current value and jump to tuning wizard.
- Filament switch sensor states.
- Filament motion sensor states.
- Runout-pause macro association shown when known.
- Optional Spoolman active spool selection, remaining weight/length, vendor, material, color and lot.
- Spool low-material warning based on file estimate plus user safety margin.

- Manual spool override when Spoolman is not present.
- Extrusion command history for current session.

Data / capability sources

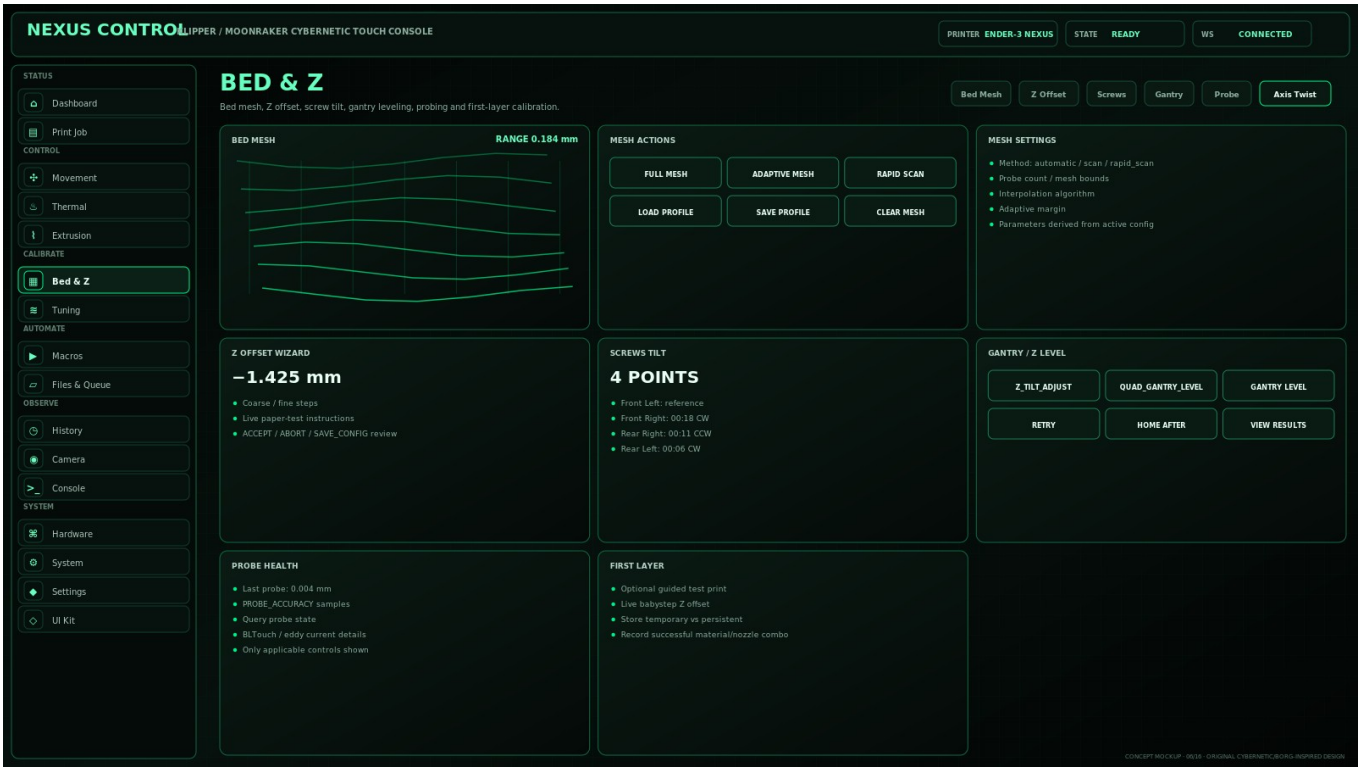
- extruder objects, gcode_move, filament_switch_sensor*, filament_motion_sensor*
- gcode_macro discovery
- Spoolman integration when present

Safety and behavior

- Large purge requires hold-to-confirm.
- Never change rotation_distance or gear_ratio from the normal extrusion page. Those belong in an explicit calibration workflow with config diff review.

Visual reference: images/05_extrusion.png and editable vector images/05_extrusion.svg.

7.6 Bed & Z



Control-plane mockup: bedz. Editable SVG is included in the package.

One calibration plane for the entire first-layer geometry stack: bed surface, probe, Z offset, screws, gantry and axis twist.

Primary sub-tabs: Bed Mesh, Z Offset, Screws, Gantry, Probe, Axis Twist.

Layout

- Large 3D/topographic bed visualization.
- Mesh actions and profiles.
- Z-offset wizard.
- Screw-tilt diagram.
- Gantry/probe tools.

Options and controls

- 3D mesh view.
- 2D topographic heat map.
- Numeric point grid.
- Display mesh range, min, max, mean and center deviation.
- BED_MESH_CALIBRATE full automatic.
- Adaptive mesh toggle and margin.
- Scan and rapid_scan methods only when applicable.
- Manual method where applicable.
- Mesh bounds and probe count shown from active config; Advanced users may override for one run.
- Save named mesh profile.
- Load named profile.
- Remove profile with confirmation.
- Clear active mesh.
- Z Offset wizard using PROBE_CALIBRATE or the applicable calibration command.
- Coarse/fine microstep buttons and numeric feedback.
- Accept / Abort controls for manual_probe state.
- SAVE_CONFIG is a separate final review step and never automatic.
- SCREWS_TILT_CALCULATE with diagram, CW/CCW direction and turn amount.

- BED_SCREWS_ADJUST for manual bed-screw workflow when configured.
- Z_TILT_ADJUST when z_tilt exists.
- QUAD_GANTRY_LEVEL when quad_gantry_level exists.
- Probe accuracy test and standard deviation.
- Probe query state.
- BLTouch / eddy-current specific status only if detected.
- Axis twist compensation calibration entry point when configured/supported.
- First-layer assisted test print with live babystep adjustment.
- Store material/nozzle-specific first-layer notes in UI settings.

Data / capability sources

- bed_mesh status object and Klipper bed mesh commands
- manual_probe, probe, configfile
- feature-gated G-code commands: BED_MESH_*, PROBE_CALIBRATE, Z_ENDSTOP_CALIBRATE, SCREWS_TILT_CALCULATE, BED_SCREWS_ADJUST, Z_TILT_ADJUST, QUAD_GANTRY_LEVEL, PROBE_ACCURACY, axis twist commands when available

Safety and behavior

- Every calibration that can write printer.cfg shows a config diff / values-to-save confirmation.
- Do not show buttons for modules not loaded in printer.objects.list / gcode help.

Visual reference: images/06_bedz.png and editable vector images/06_bedz.svg.

7.7 Tuning



Control-plane mockup: tuning. Editable SVG is included in the package.

Advanced tuning wizards that collect evidence, show recommendations, apply temporary values first and require review before persistence.

Primary sub-tabs: Input Shaper, Pressure Advance, PID, Motion, Extrusion, Test Patterns.

Layout

- Resonance spectrum hero chart.
- Input-shaper recommendation.
- Pressure Advance, PID and motion panels.
- Calibration history.

Options and controls

- Accelerometer / resonance capability test.
- Measure X resonance.
- Measure Y resonance.
- Measure both axes.
- Render PSD / resonance curves when data pipeline is available.
- Show current input_shaper values.
- Compare candidate shapers such as ZV, MZV, EI, 2HUMP_EI, 3HUMP_EI when Klipper tooling reports them.
- Apply input shaper temporarily.
- Review exact config values before SAVE_CONFIG / file edit.
- Pressure Advance tower workflow.
- Line-pattern workflow as an optional custom implementation.
- Record PA result by extruder/nozzle/material profile.
- PID autotune for hotend, bed and chamber heater when configured.
- Live PID temperature graph.
- Apply temporary PID parameters only if valid approach supported; otherwise show generated config values for review.
- Motion limits: max_velocity, max_accel, square_corner_velocity and currently applied overrides.
- Temporary velocity/accel limits through SET_VELOCITY_LIMIT.
- Reset temporary limits to configuration baseline.

- Extrusion rotation-distance calibration calculator in a separate wizard that never writes automatically.
- Flow/extrusion multiplier test assistant.
- Temperature tower launcher via user-configured macro or prepared file.
- Retraction test launcher.
- Calibration session history with time, firmware/config fingerprint, values, notes and result.
- Export calibration session JSON/CSV.

Data / capability sources

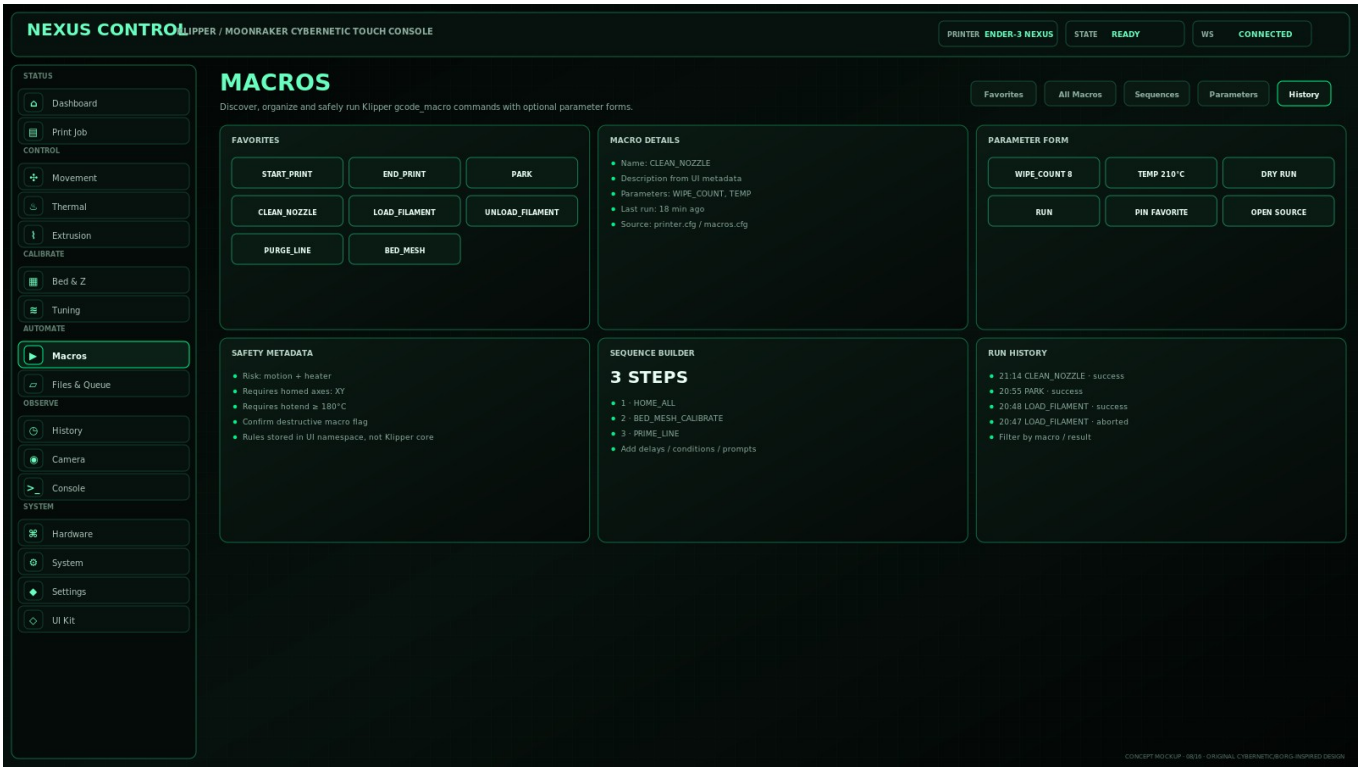
- input_shaper config/status where available
- resonance tester / accelerometer commands and generated data files when configured
- extruder pressure_advance fields
- SET_PRESSURE_ADVANCE, SET_INPUT_SHAPER, SET_VELOCITY_LIMIT and tuning commands
- file manager for generated data where applicable

Safety and behavior

- No automatic permanent config mutation. Always temporary → compare → explicit config diff → user acceptance.
- During resonance tests show a warning that the machine will move rapidly and require a clear build area.

Visual reference: images/07_tuning.png and editable vector images/07_tuning.svg.

7.8 Macros



Control-plane mockup: macros. Editable SVG is included in the package.

Turn Klipper macros into a discoverable, touch-safe command library rather than a wall of anonymous buttons.

Primary sub-tabs: Favorites, All Macros, Sequences, Parameters, History.

Layout

- Favorite macro grid.
- Macro catalog with categories/search.
- Parameter form.
- Sequence builder.
- Run history.

Options and controls

- Discover gcode_macro objects automatically.
- Friendly-name aliases stored in UI settings.
- Description field stored in UI metadata.
- Categories: Print, Filament, Calibration, Maintenance, Lighting, Custom.
- Favorite/pin macros.
- Search macros.
- Hide internal macros beginning with underscore by default; Advanced can reveal.
- Parameter form definitions stored in UI metadata: text, number, select, boolean and temperature.
- Default parameter values.
- Parameter min/max validation.
- Macro safety tags: heater, motion, homing, destructive, long-running.
- Per-macro confirmation policy.
- Per-macro required state policy, such as “must be homed” or “must be idle.”
- Run macro.
- Dry-run only where a macro-specific simulator exists; never fake dry-run execution.
- Open source config file / jump to macro definition.
- Sequence builder that chains macros with optional delays and user prompts.
- Sequence abort button.

- Session macro history and result.
- Favorite macro quick bar on dashboard.
- Import/export UI macro metadata.

Data / capability sources

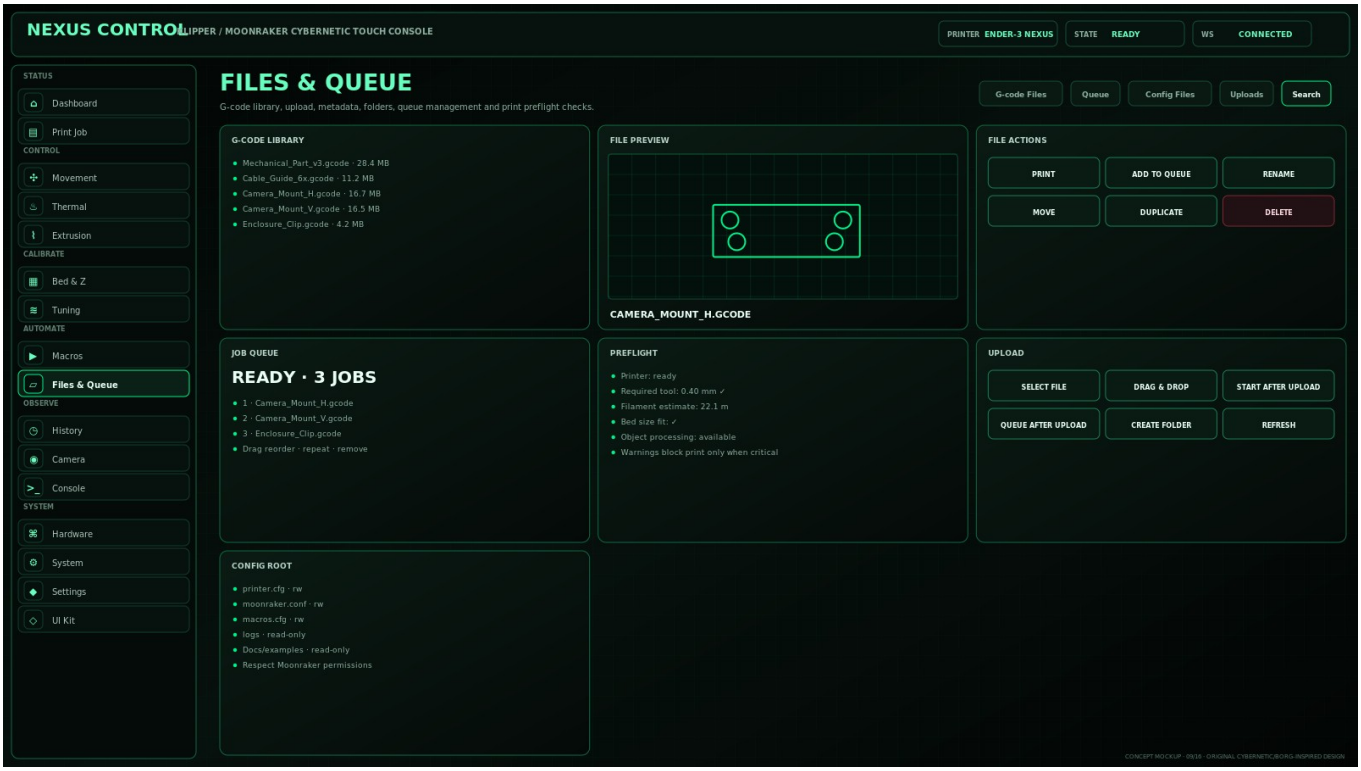
- gcode_macro* printer objects, configfile object, printer.gcode.script, Moonraker database namespace for UI metadata

Safety and behavior

- Unknown macros default to normal confirmation if they contain obvious restart/power keywords; user can explicitly classify.
- Never parse arbitrary macro Jinja and claim certainty about side effects.

Visual reference: images/08_macros.png and editable vector images/08_macros.svg.

7.9 Files & Queue



Control-plane mockup: files. Editable SVG is included in the package.

A visual file manager and job queue designed for touch, with rich metadata and clear permission boundaries.

Primary sub-tabs: G-code Files, Queue, Config Files, Uploads, Search.

Layout

- File list/grid.
- Selected-file preview.
- Metadata/preflight.
- Queue.
- Upload and config-root tabs.

Options and controls

- Browse gcodes root and subfolders.
- List or card view.
- Search filename.
- Sort by name, modified date, size, estimated duration and last print.
- Filter by material/nozzle tags when metadata exists.
- Upload G-code.
- Drag-and-drop upload on desktop.
- Start after upload.
- Queue after upload.
- Create folder.
- Rename file.
- Move file.
- Duplicate file.
- Delete file with confirmation.
- Download file.
- Show thumbnail(s).
- Show metadata and object dimensions.
- Preview toolpath / object outline.

- Start print.
- Add one or multiple files to Moonraker job queue.
- Reorder queue in UI by remove/re-add if native queue ordering does not support direct arbitrary reorder; preserve documented semantics.
- Remove queued item.
- Pause/ready queue state.
- Repeat a job by allowing duplicate queue entries.
- Config Files tab for config root with read/write permission badge.
- Read-only roots such as logs/docs/config_examples rendered accordingly.
- Config file editing belongs to an editor modal with diff, syntax display, backup and explicit save.
- Create support zip using Moonraker file ZIP API where applicable.
- Refresh metadata.

Data / capability sources

- Moonraker file manager roots, metadata and transfer APIs
- job_queue APIs
- history to show last print / outcome
- Moonraker permissions

Safety and behavior

- Never offer write/delete on a read-only root.
- Deleting a file used by an active job is blocked or clearly warned according to current server behavior.
- Config save has diff + backup policy.

Visual reference: images/09_files.png and editable vector images/09_files.svg.

7.10 History



Control-plane mockup: history. Editable SVG is included in the package.

Make printer use measurable: job outcomes, time, material, interruptions and annotations.

Primary sub-tabs: Jobs, Analytics, Materials, Failures, Totals.

Layout

- 14/30/90-day trend chart.
- Success-rate and material cards.
- Recent jobs.
- Job detail.
- Filters/exports.

Options and controls

- List Moonraker job history with pagination.
- Statuses: completed, error, cancelled, interrupted and in-progress.
- Show user that started job when present.
- Show start/end time, print_duration and total_duration.
- Show filament_used and metadata snapshot.
- Open linked G-code only if exists=true.
- Filter date range, filename, status, material and user.
- Totals: jobs, print hours, filament used.
- Success rate as a UI-derived statistic with clear denominator.
- Daily/weekly print time trend.
- Material consumption by material when spool/file metadata allows it.
- Failure annotations stored in UI namespace.
- Failure cause tags: adhesion, filament, thermal, mechanical, power, user cancel, unknown.
- Attach local UI note; optional snapshot reference.
- Compare repeated runs of the same file.
- Export CSV/JSON.
- Reset Moonraker totals only under Admin/Advanced with strong confirmation; normal History never resets server history.

Data / capability sources

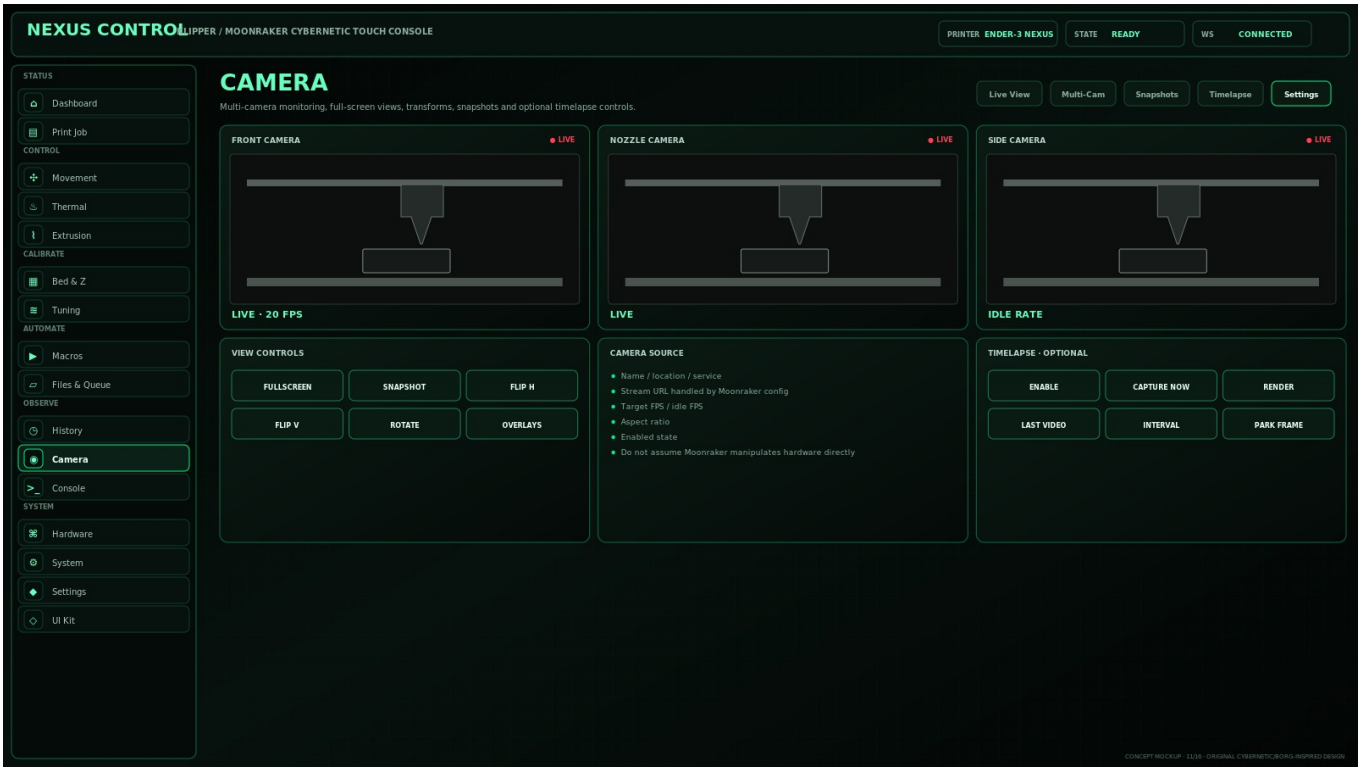
- Moonraker history API and auxiliary history fields
- file metadata
- optional Spoolman data

Safety and behavior

- Analytics must label estimates and missing data. Never invent filament/material details when metadata is absent.

Visual reference: images/10_history.png and editable vector images/10_history.svg.

7.11 Camera



Control-plane mockup: camera. Editable SVG is included in the package.

A multi-camera operator view with minimal latency, touch-friendly full screen and optional overlays.

Primary sub-tabs: Live View, Multi-Cam, Snapshots, Timelapse, Settings.

Layout

- Primary camera hero.
- Secondary camera cards.
- View controls.
- Camera configuration summary.
- Optional timelapse area.

Options and controls

- List all Moonraker webcam entries.
- Select primary camera.
- Single-camera full-screen.
- Multi-camera grid.
- Snapshot.
- Flip horizontal.
- Flip vertical.
- Rotation 0/90/180/270.
- Aspect ratio display.
- Target FPS and idle FPS display.
- Overlay print progress, temperatures, filename and time; user can disable each overlay.
- Crosshair / nozzle-zone overlay as a UI-only aid.
- Camera health: last frame time / unavailable banner.
- Auto reduce non-primary camera frame rate on constrained devices.
- Timelapse controls only when a detected integration/extension provides them.
- Snapshot gallery in UI storage only if explicitly implemented; otherwise download snapshot directly.

Data / capability sources

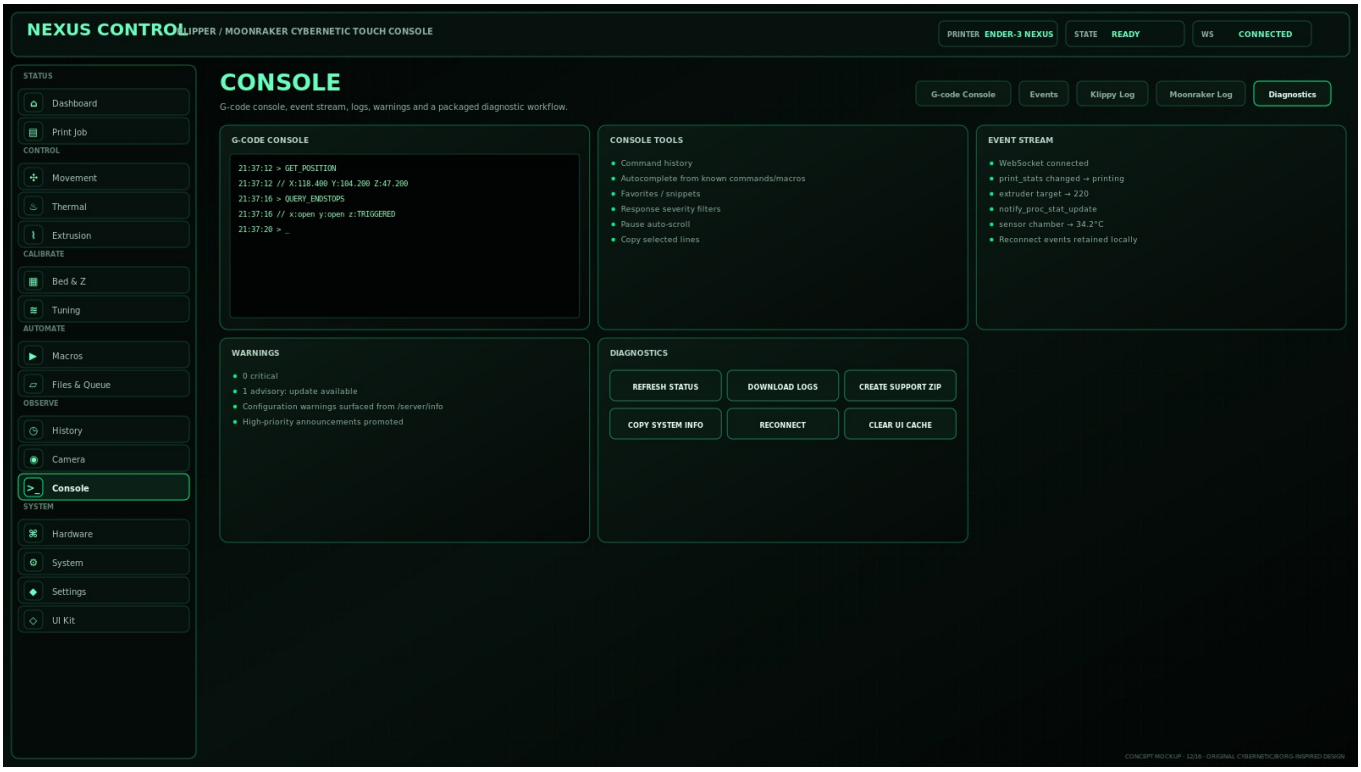
- Moonraker webcam list/item APIs; actual streams supplied by configured camera service such as crowsnest
- optional third-party timelapse extension

Safety and behavior

- Do not claim Moonraker itself controls camera hardware; it stores/shares camera configuration while an external camera service supplies the stream.

Visual reference: images/11_camera.png and editable vector images/11_camera.svg.

7.12 Console



Control-plane mockup: console. Editable SVG is included in the package.

The expert command plane plus a human-readable event and diagnostics center.

Primary sub-tabs: G-code Console, Events, Klippy Log, Moonraker Log, Diagnostics.

Layout

- Large G-code terminal.
- Command tools.
- Live event stream.
- Warnings.
- Diagnostic bundle actions.

Options and controls

- Send one or multiple G-code commands.
- Command history.
- Autocomplete from /printer/gcode/help plus discovered macros.
- Favorite snippets.
- Clear display without clearing server logs.
- Pause auto-scroll.
- Copy selection.
- Filter response lines by info/warning/error.
- Timestamp responses.
- Event stream for WebSocket connection, object-state changes, announcements and sensor notifications.
- Klippy log viewer.
- Moonraker log viewer.
- Search logs.
- Download logs.
- Create support ZIP containing selected config/log files.
- Copy system info.
- Refresh server/printer info.
- Reconnect WebSocket.

- Clear UI cache / last-known state.
- Display server warnings and failed components prominently.
- Developer pane with last JSON-RPC request/response only when Developer Mode is enabled.

Data / capability sources

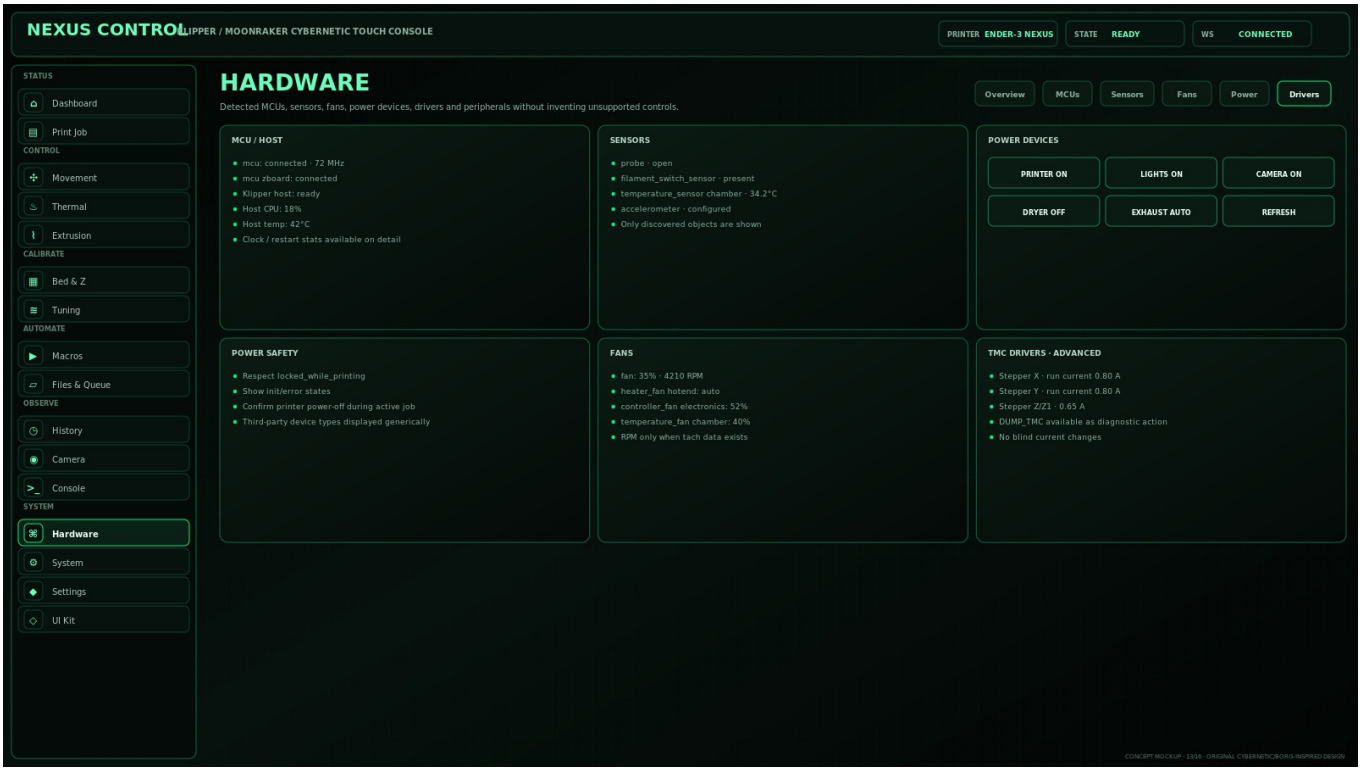
- printer.gcode.script, printer.gcode.help, server/printer info, file roots/logs, JSON-RPC notifications

Safety and behavior

- Intercept exact emergency-stop intent/M112 and invoke /printer/emergency_stop immediately.
- Console warns but does not attempt to understand all arbitrary G-code side effects.

Visual reference: images/12_console.png and editable vector images/12_console.svg.

7.13 Hardware



Control-plane mockup: hardware. Editable SVG is included in the package.

A capability-driven inventory of the machine: MCUs, sensors, fans, power devices, drivers and peripherals.

Primary sub-tabs: Overview, MCUs, Sensors, Fans, Power, Drivers, Peripherals.

Layout

- MCU/host health.
- Sensor inventory.
- Fan inventory.
- Power devices.
- Advanced TMC/peripheral diagnostics.

Options and controls

- List Klipper MCUs and versions where available.
- MCU frequency/temperature/restart stats where exposed.
- Host process/CPU/memory/temperature/network summary.
- List probe and endstop state.
- List filament sensors.
- List temperature sensors.
- List fans and RPM where configured.
- List accelerometers / resonance capability if detectable.
- List Moonraker power devices, type, state, locked_while_printing and error state.
- Turn power device on/off/toggle only when permitted.
- Batch power actions only under explicit user selection.
- Home Assistant / MQTT / HTTP / smart plug device types shown using friendly generic device cards.
- TMC driver diagnostic actions such as DUMP_TMC only for configured drivers.
- Display configured run current / hold current from config when parseable.
- Driver current editing is Advanced-only, temporary-first, and not offered when command support cannot be proven.
- List LEDs / output pins as controls only if explicitly mapped in UI metadata; do not expose arbitrary pins.
- Peripheral notes / custom labels stored in UI database.

Data / capability sources

- printer.objects.list/query
- Moonraker power device and sensor endpoints
- proc-stat notifications
- configfile
- G-code help for diagnostics

Safety and behavior

- Respect locked_while_printing from Moonraker power devices.
- Printer power-off during an active job uses a destructive confirmation and can be blocked by policy.
- Raw GPIO/pin controls are never auto-generated.

Visual reference: images/13_hardware.png and editable vector images/13_hardware.svg.

Original cybernetic/Borg-inspired visual language. No franchise logos or copied interface artwork.

7.14 System



Control-plane mockup: system. Editable SVG is included in the package.

One place for services, host health, updates, announcements, restart/shutdown and backup/support tasks.

Primary sub-tabs: Status, Updates, Services, Announcements, Storage, Backups.

Layout

- Service state.
- Host performance.
- Update cards.
- Announcements.
- Storage/backup.
- Dangerous service actions separated at the bottom.

Options and controls

- Klipper connected/state/version.
- Moonraker server status/components/warnings/failed_components.
- Host CPU usage, memory, CPU temp, network and uptime.
- Storage usage if obtainable from Moonraker machine APIs / extension.
- Update-manager status for each configured updater.
- Manual refresh updates.
- Upgrade one component.
- Upgrade all supported components only when idle.
- Show current/reported versions, channel, dirty state and warnings where provided.
- Announcements list, priority badge, open link and dismiss.
- Restart Klipper application.
- Firmware Restart.
- Restart Moonraker/system service when supported by machine APIs.
- Reboot host.
- Shutdown host.
- Emergency Stop.
- Back up config folder.

- Moonraker database backup.
- Generate support zip.
- Export UI settings.
- Restore UI settings from validated JSON.
- Show update progress/events.

Data / capability sources

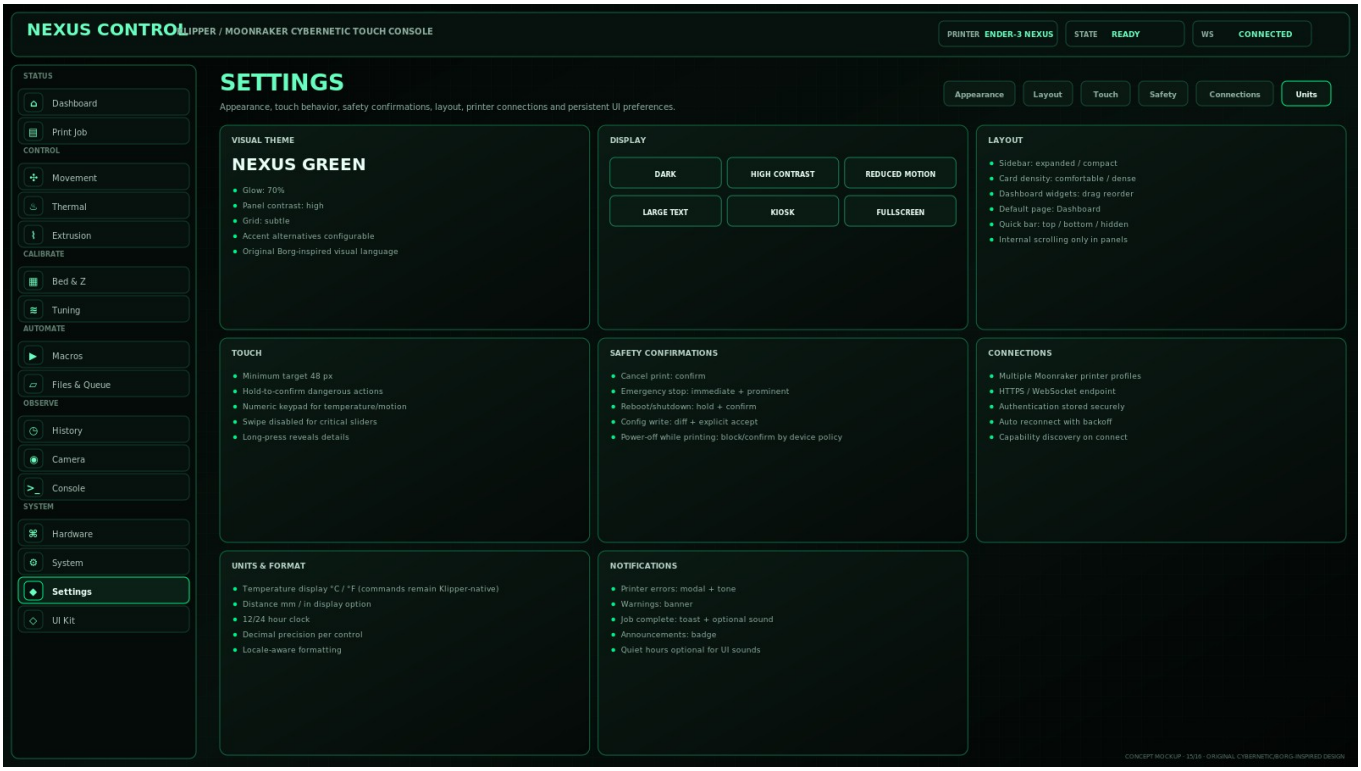
- server.info, printer.info, process-stat notifications
- update manager
- announcements
- system/machine APIs
- database backup APIs where configured
- file manager ZIP

Safety and behavior

- Update controls disabled during a print in accordance with Moonraker behavior.
- Reboot/shutdown use hold-to-confirm + second text confirmation.
- Emergency stop remains immediate, not delayed behind a normal G-code queue.

Visual reference: images/14_system.png and editable vector images/14_system.svg.

7.15 Settings



Control-plane mockup: settings. Editable SVG is included in the package.

All persistent UI preferences, connection profiles, appearance, dashboard customization, notifications and safety policy.

Primary sub-tabs: Appearance, Layout, Touch, Safety, Connections, Units, Notifications.

Layout

- Appearance and accessibility.
- Layout/widget settings.
- Touch behavior.
- Safety confirmations.
- Printer connection profiles.
- Units/localization.
- Notifications and advanced/developer preferences.

Options and controls

- Theme preset: NEXUS Green default.
- Accent choices while preserving state colors.
- Glow intensity 0–100%.
- Circuit-grid intensity 0–100%.
- Panel contrast low/normal/high.
- High Contrast mode.
- Reduced Motion mode.
- Large Text mode.
- UI density: comfortable / compact / expert-dense.
- Sidebar: expanded / compact / auto.
- Top status bar: full / compact.
- Dashboard edit mode and restore default widgets.
- Default landing page.
- Quick-action bar location.
- Touch target size: standard 48px or large 56/64px.
- Hold-to-confirm duration for selected dangerous actions.

- On-screen numeric keypad enable.
- Sound effects master toggle.
- Job-complete tone.
- Warning tone.
- Critical error tone.
- Quiet hours for UI sound.
- Kiosk mode.
- Auto full-screen preference.
- Screen dim timeout; optional wake-on-printer-event if platform supports it.
- Temperature display °C/°F.
- Distance display mm/inches while all machine commands remain mm-native.
- 12/24-hour clock.
- Locale/date format.
- Decimal precision preference for temperatures, Z offset and coordinates.
- Multi-printer connection profiles with name, endpoint and color label.
- Auto-connect last printer.
- Auto-reconnect with exponential backoff.
- Authentication/token storage through platform-appropriate secure storage.
- TLS certificate error handling with explicit user warning; no silent insecure downgrade.
- Per-printer capability cache invalidated on reconnect/version change.
- Confirmation policy for cancel, config write, power-off, reboot and shutdown.
- Emergency stop visibility: always visible / top bar + hardware-friendly mode.
- Favorite macros and metadata.
- Park positions.
- Temperature presets.
- Dashboard pinned sensors.
- Camera default and layout.
- Notification preferences.
- Developer Mode toggle.
- Raw JSON/object inspector only in Developer Mode.
- Export all UI settings JSON.
- Import UI settings with schema validation and preview.
- Reset UI settings only, never printer.cfg.
- About page with UI version, Moonraker/Klipper versions and documentation links.

Data / capability sources

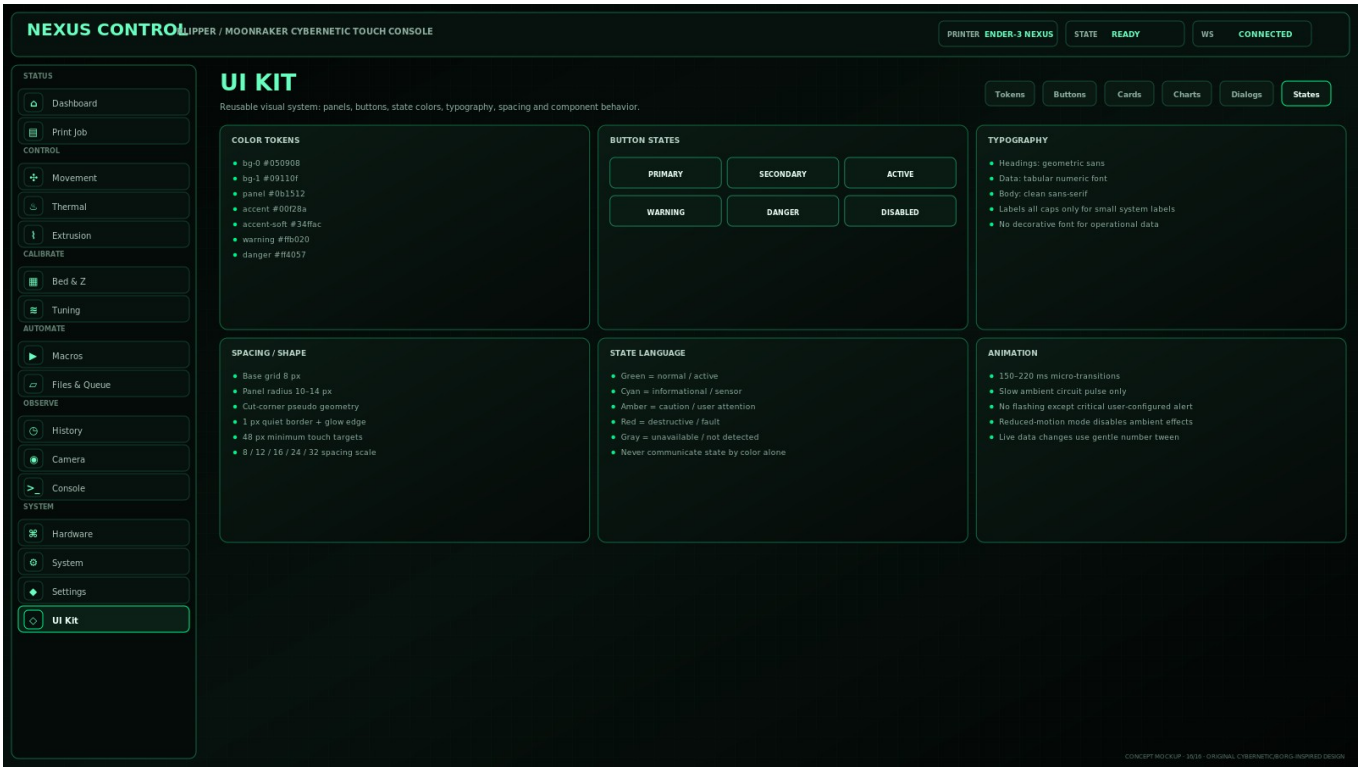
- Moonraker database namespace recommended for server-associated UI settings, plus local secure storage for secrets
- server/printer info

Safety and behavior

- Never store plaintext credentials in export files.
- A UI reset must never reset or edit printer firmware configuration.

Visual reference: images/15_settings.png and editable vector images/15_settings.svg.

7.16 UI Kit



Control-plane mockup: components. Editable SVG is included in the package.

Developer-only visual contract. This is not required as a normal operator page in production, but it is included in the package so the implementation AI can reproduce the visual language consistently.

Primary sub-tabs: Tokens, Buttons, Cards, Charts, Dialogs, States.

Layout

- Color tokens.
- Button states.
- Typography.
- Panels and charts.
- Dialog states.
- Animation rules.

Options and controls

- Primary, secondary, active, warning, destructive and disabled buttons.
- Normal, information, attention, fault and unavailable state badges.
- Metric cards with tabular numbers.
- List cards.
- Chart card.
- Camera card.
- Modal and confirmation sheet.
- Toast / banner / persistent fault rail.
- Sidebar expanded and compact components.
- Numeric keypad component.
- On-screen keyboard only for filename/search/editing needs.
- Loading skeleton, reconnect overlay and stale-data watermark.
- Tooltip / long-press help.
- Reduced-motion equivalents for every animated effect.

Data / capability sources

- No machine data. Static design-system page.

Safety and behavior

- State must never be communicated by color alone; use icon + text + color.

Visual reference: images/16_components.png and editable vector images/16_components.svg.

8. Global dialogs, drawers and overlays

- Numeric keypad: large digits, decimal, sign only when valid, Clear, Backspace, Cancel and Apply. Shows allowed min/max and current value.
- Confirmation sheet: action name, exact target, consequence, optional hold-to-confirm, Cancel as the visually dominant safe escape.
- Config diff review: before/after values, source file, line/section when known, backup checkbox, Apply and Restart requirements.
- Connection overlay: never blocks reading last-known status, but clearly watermarks values STALE and disables machine-changing controls.
- Critical fault overlay: fault message, timestamp, relevant service, Copy Details, Open Console and Emergency Stop if applicable.
- Notification center: unread announcement/warning count, filter by printer/system/job, acknowledge/dismiss where server API supports it.
- Command palette: searchable pages, macros and safe navigation actions. Destructive actions require an additional confirmation and never execute directly from fuzzy search.
- Full-screen camera: overlay controls auto-hide after a delay but a visible way to exit remains.
- Help overlay: long-press or ? explains a control in plain language without sending commands.

9. Capability gating rules

- Bed Mesh plane always exists as a navigation concept only if bed_mesh or relevant commands are present; otherwise Settings can show “Not configured.”
- Exclude Object controls require the exclude_object object and actual defined objects in the running file.
- Screws Tilt requires the configured module / command. Quad Gantry and Z Tilt are mutually independent capability cards.
- Input-shaper tuning requires the relevant Klipper commands and usually a configured accelerometer/resonance setup. Static current values may still display if available without an accelerometer.
- Spoolman sections appear only when Moonraker reports the integration.
- Webcam plane can remain available with an “Add/configure camera” empty state if Moonraker webcam component is available but no cameras exist.
- Power controls require Moonraker power devices. Never infer a relay from an arbitrary output_pin.
- Timelapse is optional and must be driven by an actual integration/extension, not an assumed Moonraker core feature.
- Update plane shows only configured update-manager entries. Upgrade buttons obey busy/printing restrictions returned by Moonraker.
- TMC advanced cards are diagnostic unless the implementation can prove supported commands/configuration for the driver.

10. Safety model

- Class A - immediate emergency: Emergency Stop. One tap/click executes the emergency endpoint immediately; optional physical-style red button UI. No second confirmation that delays the halt.
- Class B - destructive job actions: Cancel print, exclude object after printing has started, delete G-code. Confirm once with target-specific text.
- Class C - machine/service actions: firmware restart, service restart, reboot, shutdown, power off. Hold + confirm or two-step confirmation.
- Class D - persistent calibration/config writes: show temporary/test result first, then config diff, then explicit save. Create backup where possible.
- Class E - normal controls: jog, heater targets within allowed ranges, fan, macros classified as safe. Execute directly when machine state is valid.

- Disconnected/stale state disables Class B-E machine mutations. Emergency Stop remains available only if the request transport is actually reachable; otherwise show that the console cannot deliver the command.
- Do not suppress Klipper/Moonraker errors. The UI may translate them to clearer wording, but preserve raw details in an expandable section.

11. Persistence and user settings

- Use a namespaced schema such as nexus_control in Moonraker database for non-secret UI metadata that should follow the server: dashboard layout, macro labels, presets, park positions, UI tags and schema version.
- Use local secure storage / OS credential storage for secrets or authentication tokens when this is packaged as a desktop/kiosk application.
- Every persisted object includes a schema version and migration path.
- Export settings intentionally excludes secrets and connection tokens.
- Per-printer settings are keyed by a stable printer profile ID, not only hostname, so renamed hosts can be repaired without data collision.

12. Recommended implementation architecture

- Frontend: Vue 3 + TypeScript + Vite is the recommended baseline, though React/Svelte are acceptable if the existing project already uses them. Do not rewrite a working project merely to change frameworks.
- State: Pinia (Vue) or an equivalent typed store, split into connection, printer objects, files/jobs, UI settings and notifications.
- Routing: client-side routes for each control plane with the shell mounted once.
- Charts: a lightweight canvas/SVG library capable of streaming series. Avoid re-rendering the whole page for every temperature update.
- 3D bed mesh/toolpath: Three.js or a small custom WebGL/canvas renderer; provide a 2D fallback for low-power panels.
- API layer: one Moonraker client module supporting HTTP, WebSocket JSON-RPC, capability discovery, typed error normalization and command cancellation/timeouts.
- No direct filesystem assumptions from the browser. All server files go through Moonraker APIs.
- Optional desktop wrapper: Tauri or Electron only if true kiosk/system integration is needed. The core UI should remain web-deployable.

13. Suggested source tree

```
src/
  app/
    router.ts
    shell/
  api/
    moonraker-http.ts
    moonraker-ws.ts
    capabilities.ts
    commands.ts
  stores/
    connection.ts
    printer.ts
    jobs.ts
    files.ts
    cameras.ts
    system.ts
    settings.ts
  pages/
    Dashboard/ PrintJob/ Movement/ Thermal/ Extrusion/
    BedZ/ Tuning/ Macros/ FilesQueue/ History/ Camera/
    Console/ Hardware/ System/ Settings/
  components/
    nexus/ MetricCard.vue ControlButton.vue StatusBadge.vue
    NumericKeypad.vue ConfirmSheet.vue ConfigDiff.vue TrendChart.vue
```

```
  BedMeshView.vue CameraView.vue Terminal.vue FaultRail.vue
domain/
  state-normalizer.ts
  safety-policy.ts
  feature-gates.ts
  units.ts
styles/
  tokens.css
  nexus-shell.css
  components.css
```

14. Implementation phases

- Phase 0 - Inventory existing code. Do not code until the AI identifies framework, existing Moonraker client, routes, reusable components and gaps.
- Phase 1 - Build shell, tokens, responsive layout, routing, feature-gate service and reconnect/stale-state behavior.
- Phase 2 - Implement read-only live Dashboard using real server/printer/object subscriptions.
- Phase 3 - Add Print Job, Movement, Thermal and Extrusion control actions with safety policy.
- Phase 4 - Add Bed & Z and Tuning wizards. Persistent writes remain review-gated.
- Phase 5 - Add Macros, Files/Queue and History.
- Phase 6 - Add Cameras, Console/Diagnostics, Hardware and System/Updates.
- Phase 7 - Add Settings, multi-printer profiles, accessibility/reduced-motion and dashboard customization.
- Phase 8 - Performance hardening, kiosk/full-screen, reconnection torture tests, responsive QA and release packaging.

15. Acceptance criteria

- No browser-page vertical scroll at the reference 1920x1080 layout; data-heavy child panels scroll internally.
- All normal touch controls are at least 48x48 px at standard density.
- UI connects to a real Moonraker instance and the Dashboard updates without manual refresh.
- Removing an optional Klipper module from configuration removes or disables the corresponding feature cleanly after reconnect/restart.
- WebSocket disconnect produces a visible stale/reconnecting state and prevents unsafe stale-state commands.
- Pause, Resume, Cancel and Start use Moonraker print APIs and display success/error feedback.
- Emergency Stop uses the immediate Moonraker emergency-stop endpoint.
- Object exclusion works only when actual exclude_object definitions exist.
- Bed mesh renders real matrix values and supports the configured calibration methods without inventing unsupported modes.
- Config-changing wizards never persist automatically. A user must review and explicitly save.
- Power devices respect locked_while_printing.
- Updates are not offered while printing when Moonraker disallows them.
- Camera transforms and multiple cameras honor Moonraker webcam configuration.
- Screen remains readable with glow intensity at 0%.
- Reduced Motion eliminates ambient effects.
- High Contrast remains usable without relying on green/red discrimination.
- At 800x480 the same pages remain usable with compact navigation and internal scrolling.
- Every API error can be expanded to raw server detail for diagnosis.
- No copied Star Trek/Borg logos, screenshots or exact interface artwork ship with the product.

16. Package assets

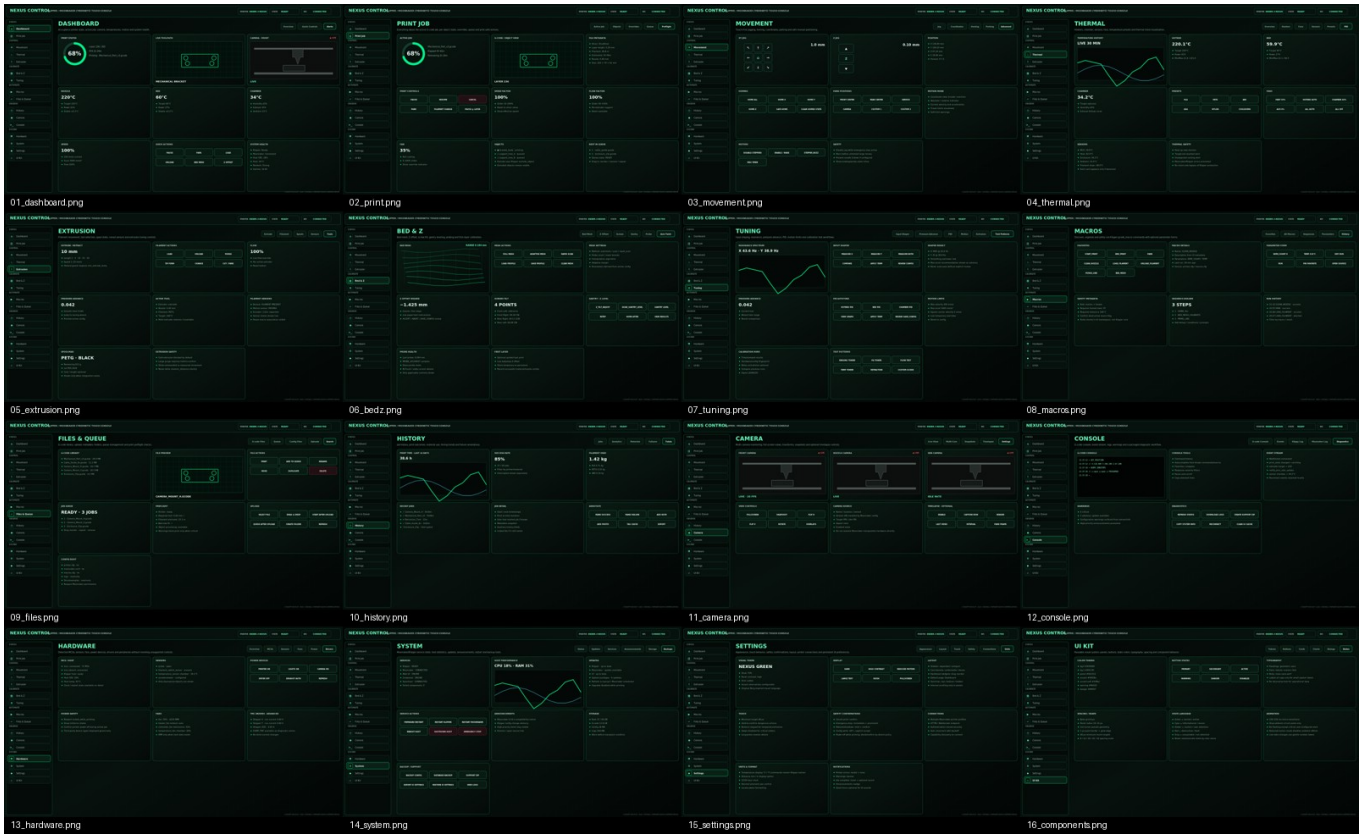
- images/00_concept_art_dashboard.png - high-fidelity visual concept art.
- images/01_dashboard.png through 16_components.png - implementation-oriented page mockups.
- Matching .svg files - editable vector page references.
- prototype/index.html - static browser prototype for style/layout exploration.
- specs/design_tokens.json - machine-readable colors, spacing, touch sizes and breakpoints.
- specs/page_manifest.json - machine-readable page/option/capability contract.
- specs/API_MAP.md - integration map and current official documentation links.
- specs/SWE2_IMPLEMENTATION_PROMPT.md - direct handoff prompt.

17. Upstream technical basis

The package was checked against current Moonraker/Klipper documentation. Moonraker provides printer status/control, file management, queue/history, webcam configuration, devices/sensors, announcements, updates, database storage and JSON-RPC notifications. Klipper exposes optional modules through printer objects and G-code commands; therefore NEXUS must discover capabilities instead of assuming them.

- https://moonraker.readthedocs.io/en/latest/external_api/printer/
- https://moonraker.readthedocs.io/en/latest/external_api/server/
- https://moonraker.readthedocs.io/en/latest/external_api/file_manager/
- https://moonraker.readthedocs.io/en/latest/external_api/job_queue/
- https://moonraker.readthedocs.io/en/latest/external_api/history/
- https://moonraker.readthedocs.io/en/latest/external_api/webcams/
- https://moonraker.readthedocs.io/en/latest/external_api/devices/
- https://moonraker.readthedocs.io/en/latest/external_api/update_manager/
- https://moonraker.readthedocs.io/en/latest/external_api/announcements/
- https://moonraker.readthedocs.io/en/latest/external_api/database/
- https://moonraker.readthedocs.io/en/latest/external_api/jsonrpc_notifications/
- https://www.klipper3d.org/Status_Reference.html
- https://www.klipper3d.org/G-Codes.html
- https://www.klipper3d.org/Bed_Mesh.html
- https://www.klipper3d.org/Exclude_Object.html
- https://www.klipper3d.org/Resonance_Compensation.html

Visual Mockup Contact Sheet



All 16 implementation-oriented control planes at a glance. Full-resolution PNG and editable SVG files are included separately.